

Jaehyeon Seo

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 Google Scholar  arXiv  GitHub  LinkedIn

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Research Interests

Extremal graph theory, graph homomorphism inequalities, independent sets and colorings

Education

Combined Master's and Doctoral Program in Mathematics

Yonsei University

Mar 2023–Present

Seoul, Korea

- Supervised by Joonkyung Lee

Graduate Study in Mathematics (not completed)

Korea Advanced Institute of Science and Technology

Mar 2020–Aug 2022

Daejeon, Korea

- Supervised by Jaehoon Kim

Bachelor's in Mathematics

Pohang University of Science and Technology

Feb 2016–Feb 2020

Pohang, Korea

- Magna Cum Laude

High school for science-gifted students

Korea Science Academy of KAIST

Feb 2013–Feb 2016

Busan, Korea

Publications and Preprints

9. **Antiferromagnetic models are clique-minimizing.** Submitted (2026). [\[arxiv\]](#)
With Joonkyung Lee.
8. **Degree-sequence bounds for independent sets via multivariate local occupancy.** Preprint (2026). [\[arxiv\]](#)
With Ewan Davies, Juspreet Singh Sandhu, Brian Tan.
7. **Lower bounds for multivariate independence polynomials and their generalisations.** Submitted (2026). [\[arxiv\]](#)
With Joonkyung Lee.
6. **Semi-Autonomous Mathematics Discovery with Gemini: A Case Study on the Erdős Problems.** Preprint (2026). [\[arxiv\]](#)
With Tony Feng, Trieu Trinh, Garrett Bingham, Jiwon Kang, Shengtong Zhang, Sang-hyun Kim, Kevin Barreto, Carl Schildkraut, Junehyuk Jung, Carlo Pagano, Yuri Chervonyi, Dawsen Hwang, Kaiying Hou, Sergei Gukov, Cheng-Chiang Tsai, Hyunwoo Choi, Youngbeom Jin, Wei-Yuan Li, Hao-An Wu, Ruey-An Shiu, Yu-Sheng Shih, Quoc V. Le, Thang Luong.
5. **Counting homomorphisms in antiferromagnetic graphs via Lorentzian polynomials.** Submitted (2025). [\[arxiv\]](#)
With Joonkyung Lee, Jaeseong Oh.
4. **Transversal cycles and paths in tournaments.** Submitted (2024). [\[arxiv\]](#)
With Debsoumya Chakraborti, Jaehoon Kim, Hyunwoo Lee.
3. **Hamilton transversals in tournaments.** *Combinatorica* **44** (2024), 1381–1400. [\[arxiv\]](#) [\[journal\]](#)
With Debsoumya Chakraborti, Jaehoon Kim, Hyunwoo Lee.
2. **On a rainbow extremal problem for color-critical graphs.** *Random Structures & Algorithms* **64** (2024), 460–489. [\[arxiv\]](#) [\[journal\]](#)
With Debsoumya Chakraborti, Jaehoon Kim, Hyunwoo Lee, Hong Liu.
1. **On 1-Subdivisions of Transitive Tournaments.** *The Electronic Journal of Combinatorics* **29**(1) (2022), P1.51. [\[arxiv\]](#) [\[journal\]](#)
With Jaehoon Kim, Hyunwoo Lee.

Research Experience and Visits

Research Visitor <i>Colorado State University</i> (Host: Ewan Davies)	May 24–31, 2026 Fort Collins, CO, US
Visiting Student Researcher <i>California Institute of Technology</i> (Host: David Conlon)	Jan 2026–Jul 2026 Pasadena, CA, US
Research Visitor <i>Shanghai Center for Mathematical Sciences</i> (Host: Yaobin Chen)	Jun 9–14, 2025 Shanghai, China
Researcher <i>Hanyang University Industry–University Cooperation Foundation</i>	Oct 2022–Feb 2023 Seoul, Korea
Visiting Student <i>Extremal Combinatorics and Probability Group, Institute for Basic Science</i>	Apr 2022–Aug 2022 Daejeon, Korea
Exchange Student <i>Graz University of Technology</i>	Feb 2018–Jul 2018 Graz, Austria

Talks

• ASIACOMB 2026 , Daejeon Convention Center <i>Degree-sequence bounds for independent sets via multivariate local occupancy</i>	Daejeon, Korea Aug 24, 2026
• Workshop for Young Mathematicians in Korea 2026 , KAIST <i>What counts do cliques minimize? Independent Sets, colorings, and more</i> (invited)	Daejeon, Korea Aug 20, 2026
• Combinatorics at the Confluence , Carnegie Mellon University <i>Clique-minimizing models</i> (poster)	Pittsburgh, PA, US Jul 20, 2026
• 2026 SIAM Conference on Discrete Mathematics , UCSD <i>Lower bounds for multivariate independence polynomials and their generalisations</i>	San Diego, CA, US Jun 23, 2026
• Flags in the Mountains , CSU Mountain Campus <i>Clique-minimizing models</i> (lightning talk)	Bellvue, CO, US Jun 4, 2026
• Graduate Student Combinatorics Conference 2026 , UIC <i>Lower bounds for multivariate independence polynomials and their generalizations</i>	Chicago, IL, US Mar 28, 2026
• Math Graduate Student Seminar , Caltech <i>Extremal questions on independent sets, colorings, and graph homomorphisms</i> (invited)	Pasadena, CA, US Feb 24, 2026
• RS&A 2025 , TU Wien <i>Counting homomorphisms in antiferromagnetic graphs via Lorentzian polynomials</i>	Vienna, Austria Aug 5, 2025
• FPSAC 2025 , Hokkaido University <i>Counting homomorphisms in antiferromagnetic graphs via Lorentzian polynomials</i> (poster)	Sapporo, Japan Jul 21, 2025
• Combinatorics Student Seminar , Shanghai Center for Mathematical Science <i>Graph homomorphism inequalities and Lorentzian polynomials</i> (invited)	Shanghai, China Jun 11, 2025
• CanadAM 2025 , University of Ottawa <i>Graph homomorphism inequalities and Lorentzian polynomials</i> (invited)	Ottawa, Canada May 21, 2025
• 4th East Asia Workshop on Extremal and Structural Graph Theory , Sun Yat-Sen University <i>Graph homomorphism inequalities and Lorentzian polynomials</i>	Guangzhou, China Mar 30, 2025
• Sinchon Seminar of Graduates 5 , Ewha Womans University <i>Graph homomorphisms and Lorentzian polynomials</i> (invited)	Seoul, Korea Dec 27, 2024
• KIAS Graduate Student Meeting in Pure Mathematics , KIAS <i>Graph homomorphisms and Lorentzian polynomials</i>	Seoul, Korea Oct 21, 2024
• 2024 Combinatorics Workshop , Chungbuk National University <i>Hamilton transversals in tournaments</i>	Cheongju, Korea Aug 29, 2024
• 2024 Korea Student Combinatorics Workshop , Gongju Hanok Village <i>Extremal graph theory and rainbow substructures</i>	Gongju, Korea Jul 30, 2024
• 2023 KMS Spring Meeting , Daejeon Convention Center <i>Rainbow oriented Hamiltonian paths and cycles in tournaments</i>	Daejeon, Korea Apr 29, 2023

- **KAIST Math Graduate Student Seminar**, KAIST
How can we find a rainbow subgraph? (invited) Online
Mar 24, 2022
- **Discrete Math Seminar**, IBS DIMAG
A rainbow Turán problem for color-critical graphs (invited) Daejeon, Korea
Jan 18, 2022
- **2021 Combinatorics Workshop**, The Bloomvista
On multicolor extremal problems Yangpyeong, Korea
Dec 21, 2021

Awards and Grants

- BK21FOUR International Joint Training Support of Outstanding Graduate Students
(4단계 BK21사업 우수대학원생 국제공동연수) Jan 2026–Jul 2026
- Best Student from the Brain Korea (BK) Evaluation in the Department Spring 2024, Spring 2025
- Excellence Award Spring Poster Presentation, School of Mathematics and Computing,
Yonsei University Spring 2024
- Dean's List POSTECH Department of Mathematics Fall 2016, Spring 2017,
Fall 2017, Spring 2019
- Excellence Prize POSTECH Programming Contest 2016
- National Scholarship (Science and Engineering) Korea Student Aid Foundation Feb 2016–Feb 2020

Teaching Experience

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|---|-------------------------------------|
| Teaching Assistant
<i>Yonsei University</i> | Mar 2023–Dec 2025
Seoul, Korea |
| • Combinatorics | Fall 2025; Fall 2024 |
| • Discrete Mathematics | Spring 2025; Spring 2024 |
| • Linear Algebra (2) | Fall 2023 |
| • Mathematics and Programming | Spring 2023 |
| Teaching Assistant
<i>Korea Advanced Institute of Science and Technology</i> | Mar 2020–Aug 2022
Daejeon, Korea |
| • Probability and Statistics, Scientific Writing (for math majors) | Spring 2022 |
| • Calculus 1, Introduction to Graph Theory | Fall 2021 |
| • Introduction to Linear Algebra, Introduction to Number Theory | Spring 2021 |
| • Differential Equations and Applications, Introduction to Differential Geometry | Fall 2020 |
| • Differential Equations and Applications | Spring 2020 |
| Excellent Student Teaching Assistant
<i>Pohang University of Science and Technology</i> | Mar 2019–Dec 2019
Pohang, Korea |
| • Calculus 2 | Fall 2019 |
| • Calculus 1 | Spring 2019 |
| Teaching Assistant
<i>KAIST Cyber Gifted Education</i> | Oct 2016–Feb 2017
Online |
| • Introduction to Number Theory, Linear Algebra | Fall 2016 |

Outreach

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| Teaching Assistant
<i>POSTECH Potential Development Camp for High School Students</i>
(POSTECH 잠재력개발과정) | Jan 2018
Pohang, Korea |
| • Tutoring high school students in undergraduate-level mathematics | |

Academic Service

- Co-organizer, Korean Student Combinatorics Workshop 2026 Winter Feb 2–6, 2026
- Co-organizer, Korean Student Combinatorics Workshop 2025 Aug 20–24, 2025
- Co-organizer, Discrete Analysis Seminar at Yonsei University Apr 2023–Present

Peer Review Activities

- SIAM Journal on Discrete Mathematics 2026
- Combinatorics, Probability and Computing; SIAM Journal on Discrete Mathematics 2025
- Journal of Graph Theory 2024

Student Seminars

- Co-organizer, Discrete Analysis & Insights Seminar at Yonsei Oct 2024–Present
Student-led seminar at Yonsei University, primarily focusing on extremal combinatorics
- Participant, Deep Seminar in Combinatorics Jan 2024–Sep 2025
Student-led seminar at KAIST
- Participant, Seminars on Gowers Norms Jul 2025–Nov 2025

Selected Academic Events

Outside of Korea

- ICM 2026 International Congress of Mathematicians, Pennsylvania Convention Center Philadelphia, PA, US. Jul 23–30, 2026
- Combinatorics at the Confluence, Carnegie Mellon University Pittsburgh, PA, US. Jul 20–22, 2026
- 2026 SIAM Conference on Discrete Mathematics, UCSD San Diego, CA, US. Jun 22–25, 2026
- Flags in the Mountains, Colorado State University Mountain Campus Bellvue, CO, US. Jun 1–5, 2026
- Graduate Student Combinatorics Conference 2026, UIC Chicago, IL, US. Mar 27–29, 2026
- RS&A 2025 Random Structures & Algorithms, TU Wien Vienna, Austria. Aug 4–8, 2025
- FPSAC 2025 Formal Power Series and Algebraic Combinatorics, Hokkaido University Sapporo, Japan. Jul 21–25, 2025
- CanaDAM 2025 Canadian Discrete and Algorithmic Mathematics Conference, University of Ottawa Ottawa, Canada. May 20–23, 2025
- 4th East Asia Workshop on Extremal and Structural Graph Theory, Sun-Yat Sen University Guangzhou, China. Mar 27–31, 2025
- Summit280, Eötvös University Budapest, Hungary. Jul 8–12, 2024
- BCC 2024 British Combinatorial Conference, Queen Mary University of London London, UK. Jul 1–5, 2024
- 3rd East Asia Workshop on Extremal and Structural Graph Theory, The Southern Beach Hotel & Resort Okinawa, Japan. Nov 1–5, 2023

AI and Mathematics

- AI and Mathematics, KIAS Seoul, Korea. Aug 3–7, 2026
Invited to the fourth week on “Algebra, Combinatorics, and Topology”
- Summer School on Formalization in Lean, ICARM Pittsburgh, PA, US. Jul 7–18, 2026
- FrontierMath: Open Problems workshop, USC Hotel Los Angeles, CA, US. Jun 9, 2026
- 2025 KIAS Winter School on Mathematics and AI, Park Roche Jeongseon, Korea. Dec 2–5, 2025
Attended team sessions on ML and Discrete Mathematics, focusing on FunSearch
- Machine Learning and Mathematics, KIAS Seoul, Korea. Jul 7–11, 2025
- AI for Mathematics Workshop on Formalization, KIAS Seoul, Korea. Dec 17, 2024
- Samsung Global Research Symposium: Mathematical Theory of AI, InterContinental Seoul COEX Seoul, Korea. Aug 1–3, 2023

Schools

- 2026 Summer School on Combinatorics and Algorithms, KAIST Daejeon, Korea. Aug 10–14, 2026
Daniel Dadush, Algorithms & Geometry of Linear Programming
Magnus Wahlström, Matroids, delta-matroids, and applications
- IBS-DIMAG Winter School on Graph Minors, IBS DIMAG Daejeon, Korea. Feb 3–19, 2025
Sebastian Wiederrecht and Maximilian Gorsky, Graph Minor Structure Theorem
- ECOPRO–CAS–SDU Joint Winter Course, IBS ECOPRO Daejeon, Korea. Jan 6–10, 2025
Huy Tuan Pham, Recent developments in probabilistic combinatorics: New synergies
- IBS–SDU Frontiers of Combinatorics, Part 2, IBS ECOPRO Daejeon, Korea. Aug 5–9, 2024
Tung Nguyen, Recent work on the Erdős–Hajnal conjecture
- 2024 Summer School on Combinatorics and Algorithms, KAIST Daejeon, Korea. Jul 22–26, 2024
Chien-Chung Huang, Combinatorial Optimization
Sebastian Wiederrecht, From treewidth to grid minor theorem
- IBS–CAS–SDU Joint Course, IBS ECOPRO Daejeon, Korea. Feb 26–Mar 1, 2024
Bjarne Schülke, Hypergraph Turán problems

In Korea

- 5th East Asia Workshop on Extremal and Structural Graph Theory, Fraser Place Namdaemun Seoul, Korea. Nov 27–30, 2025
- 1st Early Career Workshop in Extremal Combinatorics, IBS ECOPRO Daejeon, Korea. Oct 20–30, 2025
- KIAS-KAIST Workshop on current challenges in mathematics, KIAS Seoul, Korea. Aug 25–28, 2025
- Discrete Analysis at Yonsei, Yonsei University Seoul, Korea. Jun 16–20, 2025
- 2024 IBS-DIMAG Workshop on Combinatorics and Geometric Measure Theory, IBS Daejeon, Korea. Jul 14–19, 2024